called bioremediation. Bioremediation is less expensive and more sustainable than other remediations available. It is grouped into ***in situ*** bioremediation (treatment of contaminated soil or water in the site) and ***ex situ*** bioremediation (treatment of contaminated soil or water that is removed from the site and treated).

9.5.1 Microorganisms involved in bioremediation

Aerobic microbes degrade the pollutants in the presence of oxygen. They mainly degrade pesticides and hydrocarbons. *Pseudomonas putida* is a genetically engineered microorganism (GEM). Ananda Mohan Chakrabarty obtained patent for this recombinant bacterial strain. It is multi-plasmid hydrocarbon-degrading bacterium which can digest the hydrocarbons in the oil spills **(Fig. 9.4)**.

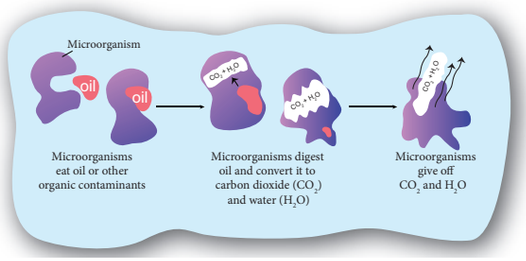


Fig. 9.4 The process of bioremediation

Nitrosomonas europaea is also capable of degrading benzene and a variety of halogenated organic compounds including trichloroethylene and vinyl chloride. *Ideonella sakaiensis* is currently tried for recycling of PET plastics **(Fig. 9.6)**. These bacteria use PETase and MHETase enzymes to breakdown PET plastic into terephthalic acid and ethylene glycol.

Anaerobic microbes degrade the pollutants in the absence of oxygen. *Dechloromonas aromatica* has the ability to degrade benzene anaerobically and to oxidize toluene and xylene. *Phanerochaete chrysosporium* an anaerobic fungus exhibits strong potential for bioremediation of pesticides, polyaromatic hydrocarbons, dyes, trinitrotoluene, cyanides, carbon tetrachloride, etc., *Dehalococcoides* species are responsible for anaerobic bioremediation of toxic trichloroethene to non-toxic ethane. *Pestalotiopsis microspora* is a species of endophytic fungus capable of breaking down and digesting polyurethane. This makes the fungus a potential candidate for bioremediation projects involving large quantities of plastics.

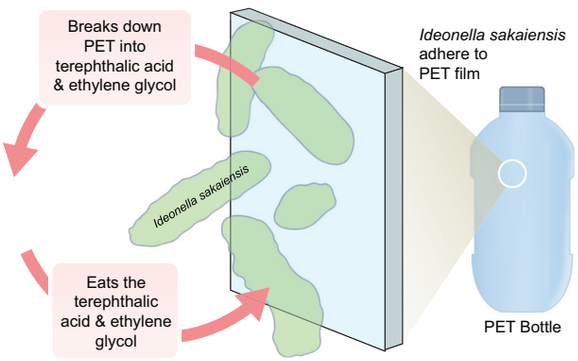


Fig. 9.6 Actions of *Ideonella sakaiensis*

Summary

All microbes are not pathogenic, many of them are beneficial to human beings. We use microbes and their derived products almost every day. Lactic acid bacteria convert milk into curd. *Saccharomyces cerevisiae* (yeast) is used in bread making. Idly and dosa are made from dough fermented by microbes. Bacteria and fungi are used in cheese making. Industrial products like lactic acid, acetic acid and alcohol are produced by microbes. Antibiotics are produced from useful microbes to kill the disease causing harmful microbes. For more than a hundred years, microbes are being used to treat sewage by the process of activated sludge formation. Bio-gas produced by microbes is used as a source of energy in rural areas. In bio-remediation naturally occurring or genetically engineered microorganisms are used to reduce or degrade pollutants.

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